E M I L Y E M E Y E R

937.623.4239 | EMEYER121@GMAIL.COM | LINKEDIN.COM/IN/EMEYER121 | GITHUB.COM/EMEYER121

EDUCATION

2021—present **Neuroscience PhD Candidate**
University of Pennsylvania
Arcaro Lab, Neuroscience Graduate Group
Interdisciplinary Training in Computational Neuroscience (ITCN) Program
Teaching Certificate from Center for Excellence in Teaching, Learning and Innovation (CETLI)

2015 – 2019 **B.S. in Mathematics and Neuroscience**
Public Health Minor
Tulane University

RELEVANT EXPERIENCE

2021 – present **Graduate Researcher**
PI: Dr. Mike Arcaro, University of Pennsylvania
- Trained and tested tree shrews on complex object recognition tasks to determine whether they have a primate-like ability to discriminate objects across identity-preserving transformations
- Adapted a front-end visual system model (ISETBio) to tree shrews, a close relative of primates, to more accurately understand the impact of optics and acuity on high-level visual behavior
- Compared object discrimination behavior to stimulus representations, from V1-like models to deep convolutional neural network activations, to probe the depth of processing required for task performance. Developed simple python module for all stimulus transformations used.
- Quantified conserved and unique features of the evolutionary expansion of primate visual cortex

2026 **Research Software Engineer Intern**
Advisor: Billy Broderick, Flatiron Institute
- Turned research code into installable, open-source software package (fenestration)
- Implemented continuous integration (CI) workflow with GitHub, wrote tests to improve code coverage, and created documentation site including API and detailed tutorials for software use cases
- Refactored existing code to improve clarity and organization for users

2019 – 2021 **Postbaccalaureate Trainee**
PI: Dr. Hendrikje Nienborg, National Eye Institute, Laboratory of Sensorimotor Research
- Utilized unsupervised clustering as a data-driven technique to identify relevant features in large-scale eye-tracking data from macaque disparity discrimination task
- Analyzed the impact of microsaccade direction and timing on behavioral outcomes, such as overall performance, bias, sensitivity, and lapse rate

Los Alamos National Lab Summer Research Intern
2019 PI: Dr. Garrett Kenyon
- Investigated significance of lateral inhibition between neurons on sparse coding model performance

2018 PIs: Dr. Nick Hengartner and Dr. Ben McMahon
- Compared models of veteran suicide risk factors in collaboration with Veterans Affairs

2017 – 2019 **Neuroscience Undergraduate Researcher**
PI: Dr. Laura Schrader, Tulane University
- Studied function of neural protein *Shox2* and its role as a thalamic pacemaker by investigating gene and protein expression localization to classify possible epilepsy therapeutic targets
- Used immunohistochemistry to label cell types with apoptotic signaling in *Shox2* knockout mice

2016 – 2019 **Epidemic Modeling Undergraduate Researcher**
PI: Dr. Mac Hyman, Tulane University
- Analyzed a risk-based model for vector-borne diseases that incorporated behavior change factors
- Developed a model for Honors Thesis project to quantify the impact of controlling diabetes on tuberculosis progression and drug resistance

PROFESSIONAL DEVELOPMENT

2023 – present Mentor for undergraduate researchers in the Arcaro lab
2023 – present ULAR-Approved Certified Trainer – Tree Shrews
2023 – present Volunteer for NGG's Action Against Bias (AAB) Application Support Program
2024 – 2026 Treasurer for Graduate-Led Initiatives and Activities (GLIA) within Neuroscience Graduate Group (NGG)
2024, 2025 Volunteer for MindCORE's Diversity & Equity Initiative (DivE In) Mentorship Weekend
2024, 2025 Teaching Assistant for UPenn Undergraduate Course – NRSC 1110
2025 Flatiron Center for Computational Neuroscience Software Workshop on Neural Data Analysis
2022-2025 UPenn Computational Neuroscience Initiative (CNI) +/- Journal Club Organizer
2022-2025 NGG Annual Retreat Organizer
2024 Volunteer mentor for Science Olympiad with G.W. Carver High School
2023 Cold Spring Harbor Course – Vision: A Platform for Linking Circuits, Behavior, and Perception
2020 Neuromatch Academy: Interactive Track

HONORS AND DISTINCTIONS

2025 UPenn Neuroscience Graduate Group Jameson-Hurvich Travel Award
2023 National Eye Institute Early Career Scientist Travel Grant for Vision Sciences Society Meeting
2023 Tree Shrew Discovery Travel Grant
2015 – 2019 Tulane University Dean's Honor Scholarship: Four-year, full-tuition scholarship
2015 – 2019 David R. Filo Scholarship: Endowed scholarship for Tulane School of Science and Engineering
2015 – 2019 Tulane Scholars Program

PUBLICATIONS

Meyer EE, Ong WS, Song C, Cottaris NP, Zhang L, Collina J, Brainard DH, Arcaro MJ. (2026) Transformation-tolerant object recognition in tree shrews despite lacking a fovea. JNeuro. *In revision.*

Meyer EE, Martynek M, Kastner S, Livingstone MS, Arcaro MJ. (2025) Expansion of a conserved architecture drives the evolution of the primate visual cortex. *Proc. Natl. Acad. Sci. U.S.A.* doi: 10.1073/pnas.2421585122

Nienborg H & Meyer EE. (2021) Neuroscience needs behavior: inferring psychophysical strategy trial by trial. *Neuron*. doi: 10.1016/j.neuron.2021.01.025

Teti M, Meyer EE, Kenyon G. (2020). Can lateral inhibition via sparse coding help explain V1 neuronal responses to natural stimuli? *IEEE Southwest Symposium on Image Analysis and Interpretation (SSIAI)*, doi: 10.1109/SSIAI49293.2020.9094598

Yu D, Febbo IG, Maroteaux MJ, Wang H, Song Y, Han X, Sun C, Meyer EE, Rowe S, Chen Y, Canavier CC, Schrader LA (2021). The transcription factor Shox2 shapes neuron firing properties and suppresses seizures by regulation of key ion channels in thalamocortical neurons. *Cereb Cortex*. doi: 10.1093/cercor/bhaa414

PRESENTATIONS

Meyer EE, Zhang L, Cottaris NP, Brainard DH, Arcaro MJ. (2025) Modeling tree shrew high-level visual behaviors. *Vision Sciences Society Annual Meeting 2025.* (Poster presentation)

Meyer EE, Arcaro MJ. Investigating the capacity for tree shrew high-level vision. *Tree Shrew Discovery 2025*. (Oral presentation)

Meyer EE, Song C, Ong WS, Arcaro MJ. Investigating the capacity for tree shrew high-level vision. *Society for Neuroscience Annual Meeting 2023*. (Poster presentation)

Meyer EE, Martynek M, Kastner S, Livingstone MS, Arcaro MJ. Larger area size, not increased number, better explains expansion of human visual cortex. *Vision Sciences Society Annual Meeting 2023*. (Poster presentation)

Meyer EE, Ong WS, Balboa M, Arcaro MJ. Assessing tree shrew high-level visual behavior using conventional and natural paradigms. *Tree Shrew Discovery 2023*. (Poster presentation)

Meyer EE, Ong WS, Balboa M, Arcaro MJ. Assessing tree shrew high-level visual behavior using conventional and natural paradigms. *Society for Neuroscience Annual Meeting 2022.* (Poster presentation)

Meyer EE, Bellet ME, Nienborg H. An unsupervised characterization of eye measurements linked to behavior emphasizes microsaccades. *NIH Postbac Poster Day 2021.* (Poster presentation)

Meyer EE, Bellet ME, Nienborg H. Using t-SNE to dissect eye movement patterns in learning and performance of a visual discrimination task. *NIH Postbac Poster Day 2020.* (Poster presentation)

Meyer EE, Rajan L, Hyman JM. (2018). Mathematical model for quantifying the impact of controlling diabetes on the spread of tuberculosis. *American Society of Tropical Medicine and Hygiene Conference.* (Poster presentation)

Meyer EE, Yu D, Schrader L. (2018) Short stature homeobox gene 2 (Shox2) is important for cell survival in mouse midline thalamus. *Tulane School of Science and Engineering (SSE) Research Day.* (Poster presentation)

Meyer EE, Rajan L, Hyman JM. (2018). Modeling the effect of chronic disease on tuberculosis progression and drug resistance. *Scientific Computing around Louisiana (SCALA) Conference.* (Poster presentation)

Meyer EE, Stafford E, Hyman JM. (2017). Risk-based model for vector-borne diseases with host behavior change. *Scientific Computing around Louisiana (SCALA) Conference.* (Oral presentation)

Meyer EE, Stafford E, Hyman JM. (2016). Modeling the 2014-2015 chikungunya outbreak. *The 8th Annual Undergraduate Research Conference at the Interface of Biology and Mathematics, sponsored by NIMBioS.* (Oral presentation)